VOLKSWAGEN

AKTIENGESELLSCHAFT

Quality management agreement between the companies of the VOLKSWAGEN GROUP and its suppliers

Formula Q-concrete contains the specifications of the companies of the VOLKSWAGEN GROUP as agreed in the contract that serve to guarantee the quality of parts and processes in the procurement and supply chain.

September 2008

Version 4, completely revised

Version 1 – February 1993

Version 2, revised – February 1994

Version 3, revised – September 1998

Version 4, completely revised – September 2008

Published by: Volkswagen AG

Group Quality Assurance Procurement (K-GQS-4)

Brieffach 1466/0, 38448 Wolfsburg

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Ladies and Gentlemen,

Rising customer requirements, global competition and pricing pressures demand products that are ready for series production and robust production processes. We have to face this challenge together if we want to achieve market success for our products.

This brochure provides a completely revised version of formula Q-concrete, which contains the basis quality requirements that you as a supplier of production materials must meet. Formula Q-concrete and other applicable documents form the basis for all enquiries and therefore must be taken into account when tenders are submitted.

In order to ensure a successful future cooperation, it is essential that the requirements described in this contract are met within the supply chain and that all suppliers provide open communication and meet the cost targets and deadlines that are set.

As a potential supplier, you agree when submitting your tender that you have read and understood the requirements described in formula Q-concrete, that you recognise, unreservedly respect and comply with these requirements and that these will be implemented in your supply chain.

You can view formula Q-concrete and any updates on the Internet at the following address: www.vwgroupsupply.com

Wolfsburg, September 2008

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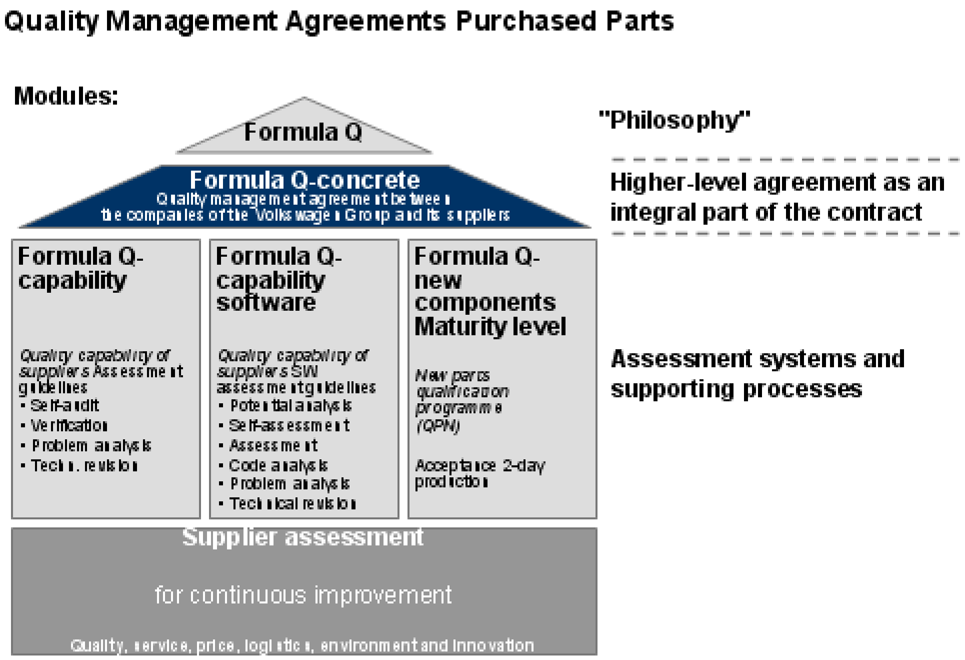

Applicable documents as amended

Group quality documents at www.vwgroupsupply.com:

- Latest information on requests for defective parts from the field for the brand VW PKW
- Formula Q-capability
- Formula Q-capability software
- Formula Q-new parts integral
- Form TF/Parts dispatch control
- Manual for original parts suppliers
- Group basis requirements software
- Group guideline product audit wiring harnesses
- Group packaging guidelines (sets and vehicle)
- Supplier manual for prototypes
- Q-certificate for pre-series production phase
- QS-guidelines logistics and CKD
- Framework agreement design management
- Managing costs for field and 0km complaints caused by suppliers
- Quality capability target agreement

Group standards directory at www.vwgroupsupply.com (FE online standards):

- High-strength connection elements, screws and bolts (VW 60250)
- Machine capability testing for measurable characteristics (VW 10130)
- Process capability testing for measurable characteristics (VW 10131)
- Accuracy – Measuring accuracy in testing processes (VW 10119)
- Overall requirements for providing service as part of the development of components (VW 99000)

VDA Books and automotive standards on www.vda-qmc.de:

- VDA Books "Quality Management in the Motor Industry" [Qualitätsmanagement in der Automobilindustrie] as well as "The Common Quality Management in the Supply Chain" [Das gemeinsame Qualitätsmanagement in der Lieferkette] (German Motor Industry Association (VDA) Series)
- ISO/TS 16949, alternatively VDA 6.1

Group environmental and social sustainability documents at www.vwgroupsupply.com

These are also supplemented by the technical delivery specifications and VW standards for each product, for example the minimum durability of components and statutory regulations, requirements, etc.

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1. Requests and preparation of tenders

1.1 Tender requirements

Before a tender is prepared, suppliers must enter their full details via the www.vwgroupsupply.com website for all production and development sites. If any information is missing, the supplier in question will not be included in the tender process.

1.2 Tender documents

The contractor must check all requirements of the tender documents for completeness, absence of contradictions, feasibility and technical state of the art (see also VW 99000, section 2.5) and must notify any discrepancies in writing (escalation).

If any amendments/additions are required, for example with regards to paint quality, residual dirt, safety equipment, airbags, etc., must be clarified with the respective technical departments before the tender is prepared (e.g. product and process requirements) and must be documented in writing.

For certain product groups, such as high-strength screws and cable sets, special Group requirements must be met. These are published on the "B2B" platform and are included with each enquiry.

1.3 Supplier concept creation

Three categories of development level are defined in the companies of the Volkswagen Group:

- Concept developer,
- SOP developer and
- Approved supplier

Depending on the development level applied, the supplier must prepare a tender in consultation with the buyer's respective technical department that contains the following minimum elements. This must be presented by the supplier when requested by the respective quality officer (e.g. quality assurance purchased parts) before the tender process.

- Description of structural design (e.g. geometry, materials, functions, software)
- Planned project organisation with the relevant contacts for the development and production sites
- Explanation of planned production processes, factory layout and supply chains
- Explanation of project schedule for target forecasts for parts approval
- Detailed description of testing and approval planning (production chain including buyer)
- Explanation of subcontractor and change management
- Explanation of measures for achieving quality targets
- Plausibility and agreement of targets (0km and field)
- Approval of target costs, deadlines, unit quantities
- Risk assessment of deadlines, costs and quality
- Definition of cost unit for chargeable special measures

- Binding feasibility statement based on specifications (see VDA maturity level assurance for new parts, RF 2 and ISO/TS 16949 section 7.2.2.2)
- Logistics and packaging concept

Any further discrepancies with regard to the requirements must be notified to the quality assurance officer of the respective company of VOLKSWAGEN GROUP in writing and escalated if necessary (gap analysis).

1.4 Quality framework agreement

The companies of the VOLKSWAGEN GROUP generally apply a zero error strategy. In addition to the supply contract regulations on defective and other liability claims, the supplier must agree a quality improvement programme with the respective quality assurance officer in writing if faults occur. If no written agreement exists, the supplier shall be required to halve any faults that occur every year.

The supplier is responsible to the companies of the VOLKSWAGEN GROUP for the supplied quality of components/modules/systems. The supplier shall manage and coordinate the subcontractors in the production and supply chain. The supplier shall apply appropriate contractual regulations to ensure that the documents applicable between the group companies and the supplier are also taken into consideration with regards to the subcontractors in the production and supply chain.

If one of the group companies specifies (sub) suppliers, so-called standard parts suppliers, within the production and supply chain for a component/module/system, these standard parts suppliers must conclude a quality framework agreement with their direct customers (i.e. the buying suppliers in the production chain). The 1st tier supplier still bears full responsibility for the quality supplied to the companies of the VOLKSWAGEN GROUP in this arrangement.

2. Quality criteria for contract awards

Before a contract is awarded each supplier is assessed based on the prevention evaluation for quality capability and proven quality performance. If no rating is available, a process evaluation of the respective production sites must be carried out before the contract is awarded in the CSC process. Non-suitability (C rating) in one of the two criteria will result in the supplier being excluded from the tender process. In addition, the companies of the VOLKSWAGEN GROUP reserve to right to conduct a project-specific risk assessment, which can also result in a C rating.

The supplier must notify the certification company of a C rating (business on hold). The buying company of the VOLKSWAGEN GROUP must be involved in subsequent steps between the certification company and the supplier.

2.1 Elements of the supplier assessment

- Evidence of a fully-functioning quality management system certified by an IATF-registered certifier in accordance with ISO/TS 16949 alternatively VDA 6.1. Without this certificate, a quality capability assessment cannot be carried out for stage A.
- Assessment of process suitability according to formula Q-capability and/or formula Q-capability software. Results are expressed based on the ratings A, B and C.
- Assessment of quality performance based on experience from previous projects and delivery and field quality. Escalation to the C rating is carried out in the "Critical approved supplier" programme.
- Product-specific and project-specific risk assessment involving experts, if required.

2.2 Quality capability target agreement

In order to ensure the quality of the components/modules/systems the companies of the VOLKSWAGEN GROUP requires the supplier to product the components/modules/systems in an A-rated production site.

If the quality capability of the supplier's production site is rated "B" at the time of award, the supplier must qualify the production site in question so that it achieves an A rating. If this requirement is no met an escalation will be carried out. Details of this are set out in the nomination letter or in the quality capability target agreement.

2.3 Design management framework agreement (KVV)

The design management framework agreement, in short KVV, defines the responsibilities in place for the development of components s between the contracting company of the VOLKSWAGEN GROUP and the supplier in advance. By defining the responsibilities for component design the suppliers are given greater control over the products developed together with the group companies. The KVV consists of a design management framework agreement, which is concluded with the supplier for the sourcing of components/modules/systems and an appendix containing the design management quota agreement, in short KV-Quota, which is concluded for each component.

2. Quality criteria for contract awards

The KV-Quota is always based on the development level of the component. The following rule of thumb applies: The more structural specifications that are applied by the Volkswagen Group - the lower the KV-Quota. The fewer structural specifications that are applied by the Volkswagen Group - the higher the KV-Quota.

The KVV is primarily based on the length of contract between the group company and the supplier.

The agreed component-related KV-Quota is applied when dealing with faulty supplies delivered by the supplier, if and when an assessment of the supplier's faulty components/modules/systems (deliveries) reveal a design-related or development-based defect.

The KVV and the KV-Quota are not applied when dealing with campaigns, particularly recalls, series defects or product liability claims. The KV-Quota also do not apply if the assessment of the faulty components/modules/systems returns a "Cannot be established" result (kFf/ tnf).

3. Working with suppliers in the product design process

3.1 New parts qualification programme (QPN-RG) / Maturity level assurance / Purchased part management

This qualification programme, which is based on the VDA maturity level (QPN-RG), must be applied consistently by the supplier throughout the entire supply chain to meet the requirements for a robust production process for new parts (see also formula Q-new parts).

This method allows users of the QPN maturity level at Volkswagen to identify scopes of delivery that are critical to maturity levels and to take any necessary measures promptly. As a result, the status in the production realisation process can be assessed across a number of disciplines based on standard maturity levels at defined project milestones. The companies of the VOLKSWAGEN GROUP reserve the right to review the current project statuses at any time with suppliers/subcontractors.

Fixed rules, including the introduction of so-called "Round tables", involves the suppliers (and subcontractors if any) of maturity-critical deliveries and the customer's international organisation in the product design process as early as possible (see also VDA maturity level new parts).

3.2 Important milestones for QPN maturity levels

- RG 0 (PF): Innovation approval for series development
- RG 1 (KE): Requirement management for scope of contract award
- RG 2 (DE): Definition of supply chain / Award of contract scopes
- RG 3 (DF,P-Frei): Approval of technical specifications
- RG 4 (B-Frei) : Procurement approval for operating equipment
- RG 5 (LF-VFF) : Standard off-tool parts, Series equipment
- RG 6 (PVS-0S) : Production, process and product approval
  (2-TP pre-check with suppliers for Prio A parts)
- RG 7 (SOP-ME) : Requalification and hand-over in series and hand-over in series (2-TP with suppliers for Prio A parts)

The supplier's purchased parts must be delivered by the part supply date (TBT) for the respective pre-series (VFF, PVS, pilot series).

The quality assurance officer (Purchased parts) shall define the criteria for acceptance of 2-day production (2-TP) based on the applicable documents. If 2-TP and prototyping are successful (see also VDA Book 2) this will be the outcome of the new parts qualification programme.

4. Series quality measures

4.1 Continuous guarantee of process capability

Process capability is a measure of the quality of a process based on the specification of the products created by the process.

On-going process capability is established and guaranteed (PFU) (see also ISO/TS 16949, sections 7.5.2 and 8.ff) in accordance with VW standards 10130, 10131 and 10119 (see also VDA 4.1).

The scope of the special characteristics that are measured to determine the Cp-, and Cpk values must be defined in the FMEA for the product and the process. These documents can be viewed at any time by the buyer.

The supplier must establish, document and, on request, notify the respective quality officer of any critical attributive characteristics (e.g. from process FMEA).

A draft protection agreement must be presented and agreed with the buyer for critical attributive characteristics (may result in A and B errors). This can be achieved, for example, using multi-level protection with Poka Yoke, occupational organisation measures, line measurements, audits, 100-percent testing, etc. for the acceptance criteria zero errors in accordance with ISO/TS 16949, section 7.1.2.

At least one method/ tool must be applied as described in VDA Books 4 and 14 for each identified process or product risk (e.g. by FMEA) in order to achieve the quality targets in accordance with the state of the art.

4.2 Product safety, product liability

The companies of the VOLKSWAGEN GROUP accept the final production liability if using purchased parts and general liability for the end product, the vehicle.

The primary production liability for the purchased parts used in the end product lies with the supplier or sub-contractor. The supplier must therefore take all organisational and technically feasible and reasonable measures to increase the product safety of its parts and the parts of its subcontractors and to minimise the risks of product liability.

The supplier shall ensure and require its subcontractors to ensure that:

- A highly-developed appreciation of quality exists throughout the company,
- The require product safety is guaranteed when components are developed,
- Special consideration is given to product safety during the quality planning stage,
- The quality capability of the production processes is guaranteed and proven,
- Appropriate series quality assurance measures are taken to minimise the probability of defective products occurring,
- Defective products are identified early on in the production workflow using appropriate measures (to minimise costs/waste of added value),
- Quality data and the legally required compliance tests are documented in sufficient detail in order to prove that the products have been manufactured in accordance with all relevant laws and safety standards
- A material tracking system can be used to pinpoint the effects of any faults that occur if required,

- Detailed information and training for the relevant staff on "Product safety and product liability" and
- Similar systems with the buyer's requirements similar to formula Q-concrete, etc. are used by all subcontractors,
- A product safety officer (PSB) is appointed for each stage in the supply chain. The 1st tier PSB must be entered in the supplier database (LDB). This information must be kept up to date at all times.
- Components with a limited durability that meet special labelling requirements, particularly in accordance with the manual for original parts suppliers.

4.3 Internal audits

To be able to assess and improve the quality capability of various divisions, the supplier must carry out regular internal audits in accordance with the requirements of ISO/TS 16949 (section 8.2.2) and formula Q-capability/formula Q-capability software.

In addition, the supplier must carry out a D/TLD self-audit at least once a year based on formula Q-capability for scopes of delivery marked as D parts.

The audit results of the D/TLD- audit must be archived for at least 15 years.

The companies of the VOLKSWAGEN GROUP reserve the right to carry out a process and product audit at any time.

4.4 Control circuits

4.4.1 Complaints manager

The companies of the VOLKSWAGEN GROUP carry out comprehensive fault management. The supplier must automatically remedy any defects that occur as quickly as possible in accordance with current regulations and must notify the sustainability of the measures taken to the relevant quality officer.

The supplier must notify the respective quality officers, logistics centres and any other partners in the supply chain of any faults that occur.

A and B faults must be remedied by the supplier with immediate effect, in some cases on request (see also formula Q-capability, section 5.3) in other words at least:

- Immediate sorting/re-working of supplies at the buyer's site and
- Use of a 100% firewall to prevent further slips.

If this cannot be carried out by internal staff, the supplier must commission a third party to carry out the work.

If an A fault occurs the supplier must send a high-level representative to the relevant quality assurance department of the buying company of the VOLKSWAGEN GROUP within 24 hours, or if requested.

4.4.2 Early warning system

Complaints management allows the group company and its supplier to obtain important early warning information about new and unknown product problems.

Guidelines on handling problems at the supplier's factory:

If the significant discrepancies are noted with regard to the defect figures for the supplier's products (e.g. rejects and re-working) the supplier must immediately report to the relevant quality officer.

This also applies if defects are identified on similar products manufactured by competitors.

Guidelines on handling problems at the buying company of the VOLKSWAGEN GROUP (production hall failures):

The supplier must work through the information supplied by the buying company of the VOLKSWAGEN GROUP immediately to avoid faulty parts being sold to end customers. Consequently, the quality information on production hall failures and ppm must also be classified and evaluated regularly and promptly.

Guidelines on handling problems in the field:

The supplier must play an active role in early warning system (hotline, commercial task force, etc.), for example with resident experts on site, to assist with the early identification of field faults. The supplier is required to assist with the fault elimination process for start-ups and must report all defective parts noted by the observation group.

4.4.3 Requirement for internal field testing with reporting obligations

As part of its product monitoring obligation the supplier must also carry out a market observation for the products it sells directly and to notify the buying company of the VOLKSWAGEN GROUP of any transferable knowledge obtained as a result.

4.5 Continuous improvement process (KVP)

The supplier is required to provide evidence of a continuous improvement process in accordance with ISO/TS 16949 section 8.5. Through this process the suppliers aims to take suitable measures to reduce internal reject and re-working quotas. This information must be presented to the companies of the VOLKSWAGEN GROUP when requested.

4.6 Change management

The supplier is required to notify the buying company of the VOLKSWAGEN GROUP of all changes to the process chain (site, product, process) prior to implementation and to obtain the approval of the relevant quality officer. If a new model is required, this must be agreed with the relevant quality officer from the buying company of the VOLKSWAGEN GROUP. Failure to observe these requirements will automatically result in a C rating (business on hold, see section 2). Any additional costs incurred for the new approval process shall be borne by the supplier.

4.7 Requalification

The supplier must guarantee quality by carrying out a regular requalification of its scope of supply in accordance with ISO/TS 16949 (section 8.2.4.1) and in accordance with VDA Robust Production Processes (section 5.3.4). Any differences to the requalification content must be agreed between the supplier and the buyer. The companies of the VOLKSWAGEN GROUP require a full requalification at least every five years. Any other requirements must be agreed in writing with the buying company of the VOLKSWAGEN GROUP.

4.8 Lessons Learned

Feedback on previous and ongoing projects, e.g. from field failures, production hall failures, project management, etc. must be used as "lessons learned" for new projects/developments and must demonstrate a measurable reduction based on previous indicators for start-ups.

4.9 Additional costs incurred due to discrepancies

Additional costs incurred as a result of field and production hall failures shall be paid for by the companies of the VOLKSWAGEN GROUP based on the originator principle.

The supplier's liability in relation to field failures is calculated using a technical factor (TF) defined between the companies of the VOLKSWAGEN GROUP and its suppliers in accordance with the corresponding documentation on quality assurance for purchased parts of the respective group company (e.g. Managing costs for field and 0km complaints caused by suppliers). Failures are generally caused by process or design faults

Technical factors are defined for a specified period.

4.10 Technical review of suppliers (TRL)

In principle the companies of the VOLKSWAGEN GROUP pursue the following global aim together with TRL (see formula Q-capability):

- Guarantee of specified part and component requirements
- Review of series production on site and all other controlling activities
- Effectiveness check of corrective measures and verification of agreed quality management standards

The companies of the VOLKSWAGEN GROUP reserve the right to carry out a TRL at any time. The TRL is notified on the working day prior to execution.

The TRL is assessed using a traffic light system. A red traffic light equates to a level 2 rating in the "Critical approved supplier" programme and will be followed up immediately with a quality meeting with the quality officer of the buying company of the VOLKSWAGEN GROUP responsible for TRL with the involvement of the supplier's top management; at this stage a decision will also be made regarding the supplier's C rating (business on hold; section 2).

4.11 "Critical approved supplier" programme“

If there are serious discrepancies in quality requirements, such as standard of delivery, prototyping and red TRL assessment, the quality officer at the buying company of the VOLKSWAGEN GROUP will escalate the supplier in the "Critical approved suppliers" programme. The programme consists of four defined levels:

Level 0:	Supplier has problems
Level 1:	Supplier is unsuccessful in solving the problems
Level 2:	Supplier requires external help to guarantee supply availability, C warning
Level 3:	Supplier does not meet group quality, C rating (business on hold; section 2)

As a rule, the ratings have a time limit. Suppliers who can provide evidence of the sustainability of the measures taken are deescalated.

The group quality officer reserves the right to enforce a direct level 3 rating in exceptional circumstances.

Terms and abbreviations

0km fault	Fault that occurs when delivering or installing the part on the assembly line.
0S (Pilot series)	A pilot series (0S) indicates the time at which the first new vehicles are constructed on an assembly line under series production conditions. A first model rating of "1" is required at this point for the purchased parts. Parts pre-production for vehicles is at least 4 weeks, or at least 6 weeks for components. Parts are used in customer vehicles.
1st tier supplier	Direct supplier for a buying plant of a company of the VOLKSWAGEN GROUP
2-day production (2-TP)	The process parameters are verified as part of "2dayproduction" to ensure the start of production in terms of quality and unit quantities.
A, B, C faults	Faults are classified according to severity using the following classes: **A1** Safety risk, breakdown **A** Unacceptable, will almost certainly result in a customer complaint **B** Annoying, disturbing, customer complaints expected, discrepancy from specifications, possible disruption to the buyer's operating procedures **C** Requires improvement, customer complaints and disruption to the operating procedures expected if frequency increases
Attributive characteristics	Attributive characteristics are characteristics that only determine the number of faults: • Proportion of faulty units (e.g. 1.8% faulty screws) • Average number of faults per unit (e.g. 3 paint faults per vehicle)
B2BB	**B**usiness to **B**usiness (e-commerce platform on the Internet for communication between suppliers and the companies of the VOLKSWAGEN GROUP) Internet: www.vwgroupsupply.com Phone Germany: 0800 - 193 30 99 Phone International: +49 - 5361 - 93 30 99 E-mail: supplierintegration@vwgroupsupply.com
Special characteristics	Critical and functional product, process and test characteristics must be defined using "System FMEA Product" and in interdepartmental teams. Other special characteristics can be defined, for example from the subsequent "System FMEA Process". In addition to legal, safety, design and process-related issues, these can include fundamental customer-related aspects.
Business on hold	A supplier production site is blocked for further orders "C" rating

Terms and abbreviations

B- Frei	**B**eschaffungs- **Frei**gabe (Procurement approval)
Cp	**P**rocess **C**apability Index; This describes the distribution of a production process.
Cpk	Process Capability [corrected]; In addition to the distribution of the production process the process capability index also takes into account the median frequency distribution at the specification limits. If the Cpk value is less than the Cp value, the median distribution will lie outside of the tolerance centre and therefore the process is not centred.
DE	**D**esign- **E**ntscheid (Design Decision)
Design Freeze	Completion of the detailed design and planning approval
D-Merkmal	**D**okumentationspflichtiges **M**erkmal (characteristic requiring documentation) of a product (see also ISO/TS 16949 section 7.3.2.3)
D part	Part with a characteristic requiring documentation (see also ISO/TS 16949 section 7.3.2.3)
D/TLD	Requiring documentation/Technical guideline documentation (see formula Q-capability, section 8, page 1)
Firewall	Safety element, which prevents faulty (not OK) parts being passed on to end customers based on a define firewall- rule book.
FMEA	**F**ailure **M**ode and **E**ffect **A**nalysis; Analysis of potential failure effects and their consequences.
Gap analysis	The gap analysis identifies strategic and operational gaps by comparing the target specifications and the expected performance in the analysed area.
IATF	International **A**utomotive **T**ask **F**orce (see ISO/TS)
ISO/TS	**I**nternational **S**tandard **O**rganisation **T**echnische **S**pezifikation (Technical Specification)
KE	**K**onzept-**E**ntscheid - (Concept decision)
KFf	**K**ein **F**ehler **f**eststellbar (Cannot be established) Faulty parts for which the cause(s) of the defect are not explained.
KVP	**K**ontinuierlicher **V**erbesserungs**p**rozess (Continuous improvement process)
KVV	**K**onzept**v**erantwortungs**v**ereinbarung Design management agreement (see applicable documents)
LDB	**L**ieferanten**d**aten**b**ank Supplier database
LF	**L**aunch- **F**reigabe (Launch approval)
ME	**M**arkt**e**inführung **(Market launch)**

Terms and abbreviations

OEM	**O**riginal **E**quipment **M**anufacturer
PF	**P**roject **F**easibility
P- Frei	**P**lanungs- **Frei**gabe (Planning approval)
PFU	Continuous process capability
Poka Yoke	System for preventing unintentional mistakes
PPM	**P**arts **p**er **m**illion
"Critical approved supplier" programme	Escalation process for suppliers with unsatisfactory performance in the production hall (0km) or in the field. The process can result in the temporary blocking of suppliers for new contracts (business on hold). (See applicable documents)
PSB	**P**rodukt**s**icherheits**b**eauftragter (Product safety officer)
PVS	The **P**roduktions-**V**ersuchs-**S**erie (Production test series) is used as the last approval stage for the assembly of individual parts. Assembly must be carried out using series-style processes with series produced parts. A minimum rating of 3 is required for purchased parts.
Q	**Q**uality
QPN	**Q**ualifizierungs**p**rogramm **N**euteile (New parts qualification programme) (see formula Q-new parts integral)
QS	**Q**ualitäts**s**icherung (Quality assurance)
RG	**R**eife**g**rad (Maturity level)
RPP	**R**obust **P**roduction **P**rocess (see VDA product manufacture and supply)
SOP	**S**tart of **P**roduction
Supply Chain	Supply chain: 1st Tier to n-Tier
Task Force	Task force commissioned to resolve a problem
TBT	**T**eile**b**ereitstellungs**t**ermin (Part supply date)
TF	The **t**echnical **f**actor is calculated on the basis of a jointly agreed technical analysis and determines the proportion paid by the originator for any damage events. $$TF = \frac{\text{Suppliers for any damage events}}{\text{Originator for any damage events}}$$ (See managing costs for field and 0km complaints caused by suppliers)
TLD (catalogue)	Catalogue in which all characteristics requiring documentation are

	listed. (See ISO/TS 16949 section 7.3.2.3)
TRL	**T**echnische **R**evision **L**ieferant (Technical review of suppliers)
tnf	**T**rouble **n**ot **f**ound parts Faulty parts for which the cause(s) of the defect are not explained.
VDA	**V**erband **d**er **A**utomobilindustrie (German motor industry association)
VFF	**V**orserien-**F**reigabe-**F**ahrzeuge (Approval before series production vehicles), initial production of series tools and series equipment with documented status for early recognition of risks.
VW-Norm	Technical standard in the companies of the VOLKSWAGEN GROUP (e.g. VW 10540 manufacturer code for vehicle parts)

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